

Madhav Kaushik

DESIGN & VERIFICATION ENGINEER (FRESHER)

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PROFESSIONAL SUMMARY

Aspiring Design & Verification Engineer with a strong foundation in **RTL Design and Functional Verification**. Hands-on experience in building **UVM-based testbenches**, writing **SystemVerilog Assertions**, and verifying complex protocols like **AXI4**. Proven problem-solver with a track record of winning national-level hackathons and leading diverse teams.

EDUCATION

Maharaja Agrasen Institute of Technology, GGSIPU	Graduate: 2025
B.Tech Electronics & Communication Engineering (Minor in AIML)	CGPA: 8.56
J.L.D.A.V Public School	2020
Class XII	90.6%
J.L.D.A.V Public School	2018
Class X	89.8%

TECHNICAL SKILLS

Verification: SystemVerilog, **UVM (Universal Verification Methodology)**, SVA (Assertions), Functional Coverage, Constrained Random Verification (CRV), OOPS

RTL Design: Verilog, FSM Design, AXI4/APB Protocols, Pipelining, Static Timing Analysis (STA)

EDA Tools: **QuestaSim**, Vivado, ModelSim, Verilator, NGSpice, Tanner EDA

Languages: C/C++, Python (Scripting), Tcl, MATLAB, Perl (Basic)

EXPERIENCE

VLSI Design Trainee (RTL Design & Verification) Jan. 2025 – Apr. 2025
Ardent Computech Pvt. Ltd.

- Implemented synthesizable RTL modules and verified functionality using **Verilog** simulations
- Debugged design issues and performed waveform analysis using **ModelSim**

KEY PROJECTS

CORDIC Sine/Cosine Generator Design & Verification (Verilog, SV & UVM)

- Designed fixed-point CORDIC processor for FPGA/ASIC implementation
- Implemented pipelined algorithm; validated precision vs. area through **RTL simulation**

E-Cycle Controller Design & Verification (SystemVerilog & UVM)

- Built **UVM verification environment** (driver, monitor, scoreboard, sequences)
- Verified motor controller FSM using **Functional Coverage** and edge-case testing

Neural Network Accelerator (Verilog)

- RISC-style datapath with parallel MAC units for ML inference
- Designed control FSM and evaluated timing, throughput, and **FPGA utilization**

ACHIEVEMENTS & LEADERSHIP

- Winner – Truth Tell Hackathon (Waves 2025), Govt of India & MeitY
- Winner – IEEE Smart City Hackathon (Hardware-Software Co-design)
- Recommended by Service Selection Board (SSB) for NDA
- Head Boy, J.L.D.A.V Public School
- Captain – School Handball & Chess Teams (inter-school championships)